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1.2.3	$S_\infty = \frac{a}{1-r}$ $S_\infty = \frac{81}{1-\frac{2}{3}}$ $= 243\text{cm}$ $243\text{ cm} > 230\text{cm}$ Yes he will be able to catch the insect **OR** Showing that he can reach a distance > 230 through **substitution of a value of *n* that is ≥ 8** in the S_n formula.	✓A substitution ✓CA value of S_∞ ✓CA conclusion **OR** ✓A substitution ✓CA value of S_n ✓CA conclusion	(3)
			[14]

QUESTION 2			
2.1.1	16 ; 0 ; – 12 ; – 20 ; …… – 16 ; – 12 ; – 8 ; ……. 4 ; 4 ; …… $2a = 4$ $a = 2$ $-16 = 3a + b$ $-16 = 3(2) + b$ $b = -22$ $16 = 2 - 22 + c$ $c = 36$ $T_n = 2n^2 - 22n + 36$	✓A value of a ✓CA value of b ✓CA value of c ✓CA answer	(4)
2.1.2	32 ; 28 ; 24 ; 20 ; …… $a = 32$ $d = -4$ $T_n = a + (n-1)d$ $T_n = 32 + (n-1)(-4)$ $T_n = -4n + 36$ **Answer only: Full marks**	✓A substitution ✓CA answer	(2)
2.1.3	$2n^2 - 22n + 36 = -4n + 36$ $2n^2 - 18n = 0$ $n^2 - 9n = 0$ $n(n-9) = 0$ $n = 0$ or $n = 9$ $n = 9$ only $\therefore T_{17}$ and T_{18} are the terms	✓CA equating ✓CA factors ✓CA $n = 9$ only ✓CA answer	(4)